

Unit 3: CAD Modeling for Aerospace Components

Quiz 3A: Fusion Interface, File Setup, and CAD Workflow

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- CAD stands for...
A. Computer-Aided Design
B. Creative Art Drawing
C. Class Assignment Data
D. Controlled Aircraft Drawing
- Why should CAD files be named carefully?
A. To make them harder to find
B. To track parts, versions, and project ownership
C. To avoid saving
D. To remove documentation
- Which workflow is usually best?
A. Sketch, constrain/dimension, feature, inspect, revise
B. Print first, think later
C. Decorate, delete, export
D. Skip saving
- What does the browser/tree in CAD help show?
A. Model features, components, and history
B. Only grades
C. Only websites
D. Only color choices
- Why save versions?
A. To track changes and recover earlier work
B. To waste storage only
C. To hide mistakes
D. To avoid revisions
- A component should be used when...
A. A part is part of a larger assembly or system
B. The file is empty
C. No geometry exists
D. Only text is needed
- What is the purpose of the view cube or navigation tools?
A. To change the view/orientation of the model
B. To grade the model
C. To export quizzes
D. To make parts stronger

8. Why should students inspect a model before submitting?
- A. To check dimensions, features, and errors
 - B. To make it take longer
 - C. To skip rubrics
 - D. To delete constraints

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|------------------------------|--|
| _____ CAD | A. Software-based design and modeling. |
| _____ Browser/tree | B. The model area showing features, bodies, components, and history. |
| _____ Version control | C. Saving and tracking file changes over time. |
| _____ Component | D. A distinct part within a larger design or assembly. |
| _____ Navigation | E. Tools for orbiting, panning, zooming, and changing views. |

Part C: Short Response

1. Describe a clean CAD workflow for creating an aerospace bracket.

2. Explain why file setup matters before modeling begins.

Unit 3: CAD Modeling for Aerospace Components

Quiz 3B: Sketch Constraints, Dimensions, and Parametric Modeling

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- A sketch constraint controls...
A. Geometry relationships such as horizontal, vertical, tangent, or equal
B. The final grade
C. The printer temperature only
D. The page color
- A dimension in a CAD sketch controls...
A. Size or location numerically
B. Only the file name
C. Only the appearance of notes
D. Nothing
- Parametric modeling means...
A. Designs are controlled by dimensions and relationships that can be edited
B. Parts cannot be changed
C. Only pictures are used
D. Models are always random
- Which constraint keeps two circles the same size?
A. Equal
B. Delete
C. Color
D. Export
- Which constraint could align an edge straight across the screen?
A. Horizontal
B. Material
C. Browser
D. Render
- Why should sketches be fully defined when possible?
A. To make geometry predictable and easier to edit
B. To make the file bigger
C. To hide dimensions
D. To prevent changes forever
- What may happen if a sketch is under-constrained?
A. Geometry may move unexpectedly
B. It prints automatically
C. It becomes stronger
D. It cannot be opened

8. Why are parameters useful in aerospace design?
- A. They allow important dimensions to be adjusted consistently B. They remove accuracy
C. They replace testing D. They only change colors

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|---------------------------------------|---|
| _____ Constraint | A. A relationship that controls sketch geometry. |
| _____ Dimension | B. A numerical value controlling size or location. |
| _____ Parametric model | C. A model driven by editable dimensions and relationships. |
| _____ Fully defined sketch | D. A sketch where geometry position and size are completely controlled. |
| _____ Under-constrained sketch | E. A sketch where geometry can still move unexpectedly. |

Part C: Short Response

1. Explain why a fully defined sketch is usually better than a loose sketch.

2. Give an example of a parameter that could be useful in a payload mount design.

Unit 3: CAD Modeling for Aerospace Components

Quiz 3C: Core Features, Model Inspection, Exporting, and CAD Documentation

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- Extrude is commonly used to...
 - Create 3D geometry from a 2D profile
 - Delete all sketches
 - Name a file
 - Measure only weight
- Fillets are often used to...
 - Round edges
 - Make holes square
 - Erase dimensions
 - Export images
- A chamfer creates...
 - A beveled edge
 - A random pattern
 - A hidden line only
 - A title block
- Why inspect model dimensions?
 - To verify the model matches requirements
 - To decorate the model
 - To avoid criteria
 - To make it slower
- Which file may be used for 3D printing?
 - STL
 - TXT only
 - JPEG only
 - PDF only
- CAD documentation may include...
 - Screenshots, dimensions, feature notes, and exported files
 - Only jokes
 - Only file names
 - Only colors
- Why should a model be checked before export?
 - Errors can become fabrication problems
 - It is optional forever
 - It removes the need for saving
 - It guarantees success

8. Which feature is useful for hollowing a solid part?
- | | |
|----------|------------|
| A. Shell | B. Save As |
| C. Zoom | D. Undo |

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|----------------------|--|
| _____ Extrude | A. A feature that pulls a 2D profile into 3D geometry. |
| _____ Fillet | B. A rounded edge feature. |
| _____ Chamfer | C. A beveled edge feature. |
| _____ Shell | D. A feature that hollows or thins a solid body. |
| _____ Export | E. Saving a design into another file format for use elsewhere. |

Part C: Short Response

1. Describe two checks you should complete before exporting a CAD model.

2. Explain how poor CAD documentation can cause problems for a team.
